

Requirements Table

Hyperlocal Courier Dispatch & Tracking Engine

PES1UG24CS391

Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match an outgoing courier request with the nearest active delivery rider within a 3 km radius .	High	Pass: Rider receives a dispatch notification with the pickup location. Fail: Order is assigned to an offline rider.	Ensures courier requests are assigned quickly to a suitable nearby rider.
FR-002	Functional	The system shall optimize the delivery route when a rider has multiple pickup or delivery stops.	High	Pass: The system generates an optimized sequence of stops. Fail: Stops are assigned without route optimization.	Reduces unnecessary travel and improves delivery efficiency.
FR-003	Functional	The system shall provide the sender with the delivery rider's live GPS location and current delivery status during an active delivery.	High	Pass: Sender can view the rider's current location and delivery status. Fail: Tracking information is unavailable or significantly outdated.	Provides real-time visibility of the courier.
FR-004	Functional	The system shall verify the recipient using an OTP before marking the parcel as delivered.	High	Pass: Valid OTP allows delivery completion. Fail: Invalid OTP prevents the parcel from being marked as delivered.	Ensures that the parcel is delivered to the intended recipient.
FR-005	Functional	The system shall notify the sender when important delivery events occur, including rider assignment, parcel pickup, and delivery completion.	Medium	Pass: Sender receives the corresponding status notification. Fail: A delivery status changes without notifying the sender.	Keeps the sender informed throughout the delivery process.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Delivery status and live GPS telemetry from riders shall be transmitted to the sender client with under 2-second latency .	High	Pass: Benchmarking tests confirm the target latency and security standards under simulated peak load. Fail: The target latency or security standards are not met.	Low-latency updates are essential for effective live delivery tracking.
NFR-002	Security	The system shall protect courier tracking, delivery status, and OTP-related information from unauthorized access during transmission and access.	High	Pass: Unauthorized users cannot access protected delivery or tracking information. Fail: Protected delivery information can be accessed without authorization.	Courier location and delivery information must remain secure.